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Ambulatory infusion pumps and assemblies for use with same

Abstract

Ambulatory infusion pumps, medicament reservoirs, and medicament sealing assemblies, including a variety of trocar seal assemblies and fill plug seal assemblies, plus related components, as well as component combinations and related methods.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/943,418, filed Jul. 30, 2020, which is a divisional of U.S. application Ser. No. 15/430,513, filed Feb. 12, 2017, now U.S. Pat. No. 10,940,264, which claims the benefit of and priority to previously filed U.S. Provisional Patent Application Ser. No. 62/294,941, filed Feb. 12, 2016, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(1) The present devices and methods relate generally to ambulatory infusion pumps and seals for those pumps.

2. Description of the Related Art

(2) Ambulatory infusion pumps (also referred to herein simply as “infusion pumps”) are relatively small, at least substantially self-contained devices that are used to introduce drugs and other infusible substances (collectively “medicament”) into patients' bodies. Some infusion pumps are configured to be worn on a belt, carried in a clothing pocket, or the like. Other infusion pumps are configured to be adhered to skin in patch-like fashion. Infusion pumps are advantageous in that they may be used to, for example, subcutaneously introduce (or “infuse”) medicament on an ongoing or even continuous basis outside of a clinical environment. Infusion pumps are also advantageous in that they greatly reduce the frequency of subcutaneous access events such as needle-based shots. One example of a medicament that may be introduced by an infusion pump is a liquid formulation of insulin. Other exemplary medicaments that may be introduced by an infusion pump include, but are not limited to, drugs that treat cancers and drugs that suppress the perception of pain.

(3) Many conventional infusion pumps have improved patient health and quality of life. Nevertheless, the present inventors have determined that conventional infusion pumps are susceptible to a wide range of improvements. By way of example, but not limitation, the present inventors have determined that it would be desirable to provide an infusion pump that is smaller, simpler, and less costly than conventional infusion pumps, while also being more accurate than conventional infusion pumps.

SUMMARY

(4) An apparatus in accordance with at least one of the present inventions includes a medicament reservoir, a reservoir outlet in fluid communication with the medicament reservoir, and a trocar seal assembly, associated with the reservoir outlet and including a trocar with a rod and a sharp end that is movable relative to the reservoir outlet, configured to prevent medicament from flowing through the reservoir outlet when the trocar is in a first position and to permit medicament flow through the reservoir outlet in response to the trocar being moved from the first position. The present inventions also include infusion pumps with such apparatus.

(5) An apparatus in accordance with at least one of the present inventions includes a medicament reservoir, a reservoir outlet in fluid communication with the medicament reservoir, a fill port in fluid communication with the medicament reservoir, and a fill plug seal assembly, associated with the fill port and the reservoir outlet and including a core pin, configured to prevent medicament from flowing through the reservoir outlet when the core pin is in a first position and to permit medicament flow through the reservoir outlet in response to the core pin being moved from the first position. The present inventions also include infusion pumps with such apparatus.

(6) The features and attendant advantages of the present inventions will become apparent as the inventions become better understood by reference to the following detailed description when considered in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Detailed description of exemplary embodiments will be made with reference to the accompanying drawings.

(2) FIG. 1A is a perspective view of an exemplary infusion pump system in an assembled state.

(3) FIG. 1B is an exploded perspective view of the infusion pump system illustrated in FIG. 1A, including a durable assembly and a disposable assembly.

(4) FIG. 2 is a top view of certain components of the infusion pump system illustrated in FIGS. 1A and 1B.

(5) FIG. 2A is a schematic view showing a use of the infusion pump system illustrated in FIGS. 1A and 1B.

(6) FIG. 2B is a schematic view showing another use of the infusion pump system illustrated in FIGS. 1A and 1B.

(7) FIG. 3A is a perspective view of an exemplary durable assembly.

(8) FIG. 3B is a perspective view of certain components of the durable assembly illustrated in FIG. 3A.

(9) FIG. 4A is a perspective view of an exemplary disposable assembly.

(10) FIG. 4B is a perspective view of certain components of the disposable assembly illustrated in FIG. 4A.

(11) FIG. 5A is a perspective view of certain components of a durable assembly and a disposable assembly of an exemplary infusion pump system.

(12) FIG. 5B is a perspective view of the components of the exemplary durable assembly illustrated in FIG. 5A.

- (13) FIG. 5C is a perspective view of the components of the exemplary disposable assembly illustrated in FIG. 5A.
- (14) FIG. 6 is a perspective section view of components of the exemplary infusion pump system of FIG. 5A, revealing a gap between certain components of the durable and disposable assemblies.
- (15) FIG. 7 is a front view showing a patient's skin being cleaned.
- (16) FIG. 8 is a flow chart illustrating an exemplary disposable assembly removal and replacement method.
- (17) FIG. 9A is a perspective view of another exemplary infusion pump system in an assembled state.
- (18) FIG. 9B is a perspective view of an exemplary durable assembly of the infusion pump system of FIG. 9A.
- (19) FIG. 9C is a perspective view of certain components of the durable assembly illustrated in FIG. 9B.
- (20) FIG. 9D is a perspective view of an exemplary disposable assembly of the infusion pump system of FIG. 9A.
- (21) FIG. 9E is a perspective view of certain components of the disposable assembly illustrated in FIG. 9D.
- (22) FIG. 10A is a perspective view of certain exemplary components of a disposable assembly.
- (23) FIG. 10B is an exploded perspective view of certain components illustrated in FIG. 10A.
- (24) FIG. 11A is an exploded section view of certain exemplary components of a disposable assembly.
- (25) FIG. 11B is a section view of the components of FIG. 11A, shown in their assembled positions.
- (26) FIGS. 11C-11F are section views showing additional exemplary components for use with the components of FIG. 11A, shown in their assembled positions.
- (27) FIG. 12 is a section view showing certain alternative components of a disposable assembly.
- (28) FIG. 13A is a side view showing additional alternative components of a disposable assembly.
- (29) FIG. 13B is a section view of certain exemplary components of the assembly of FIG. 13A.
- (30) FIG. 13C is a ninety degree section view of the components of FIG. 13B.
- (31) FIG. 13D is a side view of certain exemplary components of the assembly of FIG. 13A, shown in a pre-dispensing or a storage position.
- (32) FIG. 13E is a ninety degree section view of the components of FIG. 13D, shown after the button of FIG. 13D has been pressed.
- (33) FIG. 13F is a side view of certain components of the assembly of FIG. 13D.
- (34) FIG. 13G is a section view of the components of FIG. 13F.
- (35) FIG. 13H is a section view of certain exemplary components of the assembly of FIG. 13A, shown in a pre-dispensing or a storage position.
- (36) FIG. 13J is a section view of the components of FIG. 13H, shown after the button of FIG. 13H has been pressed.
- (37) FIG. 13K is a section view of certain exemplary components of the assembly of FIG. 13A.
- (38) FIG. 13M is a close-up section view of a portion of FIG. 13K.
- (39) FIG. 13N is a section view taken along line 13N-13N in FIG. 13M.
- (40) FIG. 13P is a side view of certain exemplary components of the assembly of FIG. 13A, shown in a pre-dispensing or a storage position.
- (41) FIG. 13Q is a perspective view of certain components of the assembly of FIG. 13P.
- (42) FIG. 14A is a perspective view showing alternative components of a disposable assembly.
- (43) FIG. 14B is a partially exploded section view of certain exemplary components of the assembly of FIG. 14A.
- (44) FIG. 14C is a perspective view of certain components of the assembly of FIG. 14A.
- (45) FIG. 14D is a perspective view showing additional components of the assembly of FIG. 14A.

- (46) FIGS. 15A and 15B are side views showing alternative components of a disposable assembly.
- (47) FIGS. 16A-16C are perspective section views showing additional alternative components of a disposable assembly, shown in a pre-dispensing or a storage position.
- (48) FIG. 16D is a perspective section view of the components of FIGS. 16A-16C, shown after the button of FIG. 16C has been pressed.
- (49) FIGS. 17A-17C are perspective section views showing further alternative components of a disposable assembly, shown in a pre-dispensing or a storage position.
- (50) FIG. 17D is a perspective section view of the components of FIGS. 17A-17C, shown after the button of FIG. 17C has been pressed.
- (51) FIG. 17E is another section view of the components of FIGS. 17A-17C, shown after the button of FIG. 17C has been pressed.
- (52) FIG. 18A is a side view of certain exemplary components that may be used with a disposable assembly, shown in a pre-dispensing or a storage position.
- (53) FIG. 18B is a side view of the components of FIG. 18A, shown with an additional exemplary component and in a pre-dispensing or a storage position.
- (54) FIG. 18C is a perspective view of the components of FIG. 18B, shown in a pre-dispensing or a storage position.

DETAILED DESCRIPTION

- (55) The following is a detailed description of the best presently known modes of carrying out the inventions. This description is not to be taken in a limiting sense, but is made merely for the purpose of illustrating the general principles of the inventions.
- (56) It should also be noted here that the specification describes structures and methods that are especially well-suited for the subcutaneous delivery of high concentration insulin (i.e., U-200 insulin and above) such as U-500 insulin as well as lower concentration insulin such as U-100 insulin. Nevertheless, it should be appreciated that the present inventions are applicable to a wide variety of infusion pumps and medicaments. By way of example, but not limitation, the inventions may employ, for fluid displacement, a reservoir with a plunger, a fluid displacement device in the form of a plunger pusher, and a drive mechanism that includes a motor, or other fluid displacement devices, regardless of the type of reservoir employed, piston pumps (e.g., electromagnet pumps), MEMS pumps, peristaltic pumps and any other suitable pumps as well as corresponding drive mechanisms. Exemplary infusion pumps that include a reservoir with a plunger, a fluid displacement device in the form of a plunger pusher, and a drive mechanism are described in U.S. patent application Ser. No. 12/890,207, filed Sep. 24, 2010, and corresponding U.S. patent publication number 2012/0078170, both of which are incorporated by reference in their entireties, and in U.S. provisional patent application Ser. No. 62/057,273, filed Sep. 30, 2014, corresponding U.S. patent application Ser. No. 14/869,906, filed Sep. 29, 2015, and corresponding U.S. patent publication number 2016/0089491, each of which are incorporated by reference in their entireties, and in U.S. provisional patent application Ser. No. 62/117,565, filed Feb. 18, 2015, corresponding U.S. patent application Ser. No. 15/042,093, filed Feb. 11, 2016, and corresponding U.S. patent publication number 2016/0235913, each of which are also incorporated by reference in their entireties. The present inventions are also applicable to medicaments such as, for example, drugs to mask pain, chemotherapy and other cancer related drugs, antibiotics, hormones, GLP-1, Glucagon, various other drugs that include large molecules and proteins that may require a high level of delivery accuracy, as well as to relatively high concentration insulin (i.e., U-200 insulin and above) such as U-500 insulin, as well as lower concentration insulin, such as U-100 insulin.
- Aforementioned U.S. patent publication number 2012/0078170, U.S. provisional patent application Ser. No. 62/057,273, U.S. patent application Ser. No. 14/869,906, and corresponding U.S. patent publication number 2016/0089491 each also describe patient interaction with and use of infusion pumps such as the exemplary infusion pumps described herein.
- (57) As noted above, some ambulatory infusion pumps are intended to be worn on a belt, carried in

a pocket, or otherwise supported within a holder of some kind (referred to collectively as “pocket pumps”). Such infusion pumps transfer fluid from a reservoir to an infusion set by way of an elongate tube. Subcutaneous access may be obtained by way of a cannula in the infusion set. Other ambulatory infusion pumps are intended to be adhered to the skin above the delivery site (sometimes referred to as “patch pumps”). Here, the cannula or other subcutaneous access device may extend directly from the infusion device. Given these modes of use, patients typically prefer the device to be as small as possible so it is more comfortable, less obtrusive, and less visible. In addition, patients want a device that is easy and convenient to use.

(58) An exemplary ambulatory infusion system, which is generally represented by reference numeral **100** in FIGS. **1A**, **1B**, and **2**, includes a durable assembly **200** and a disposable assembly **300**. Exemplary durable assembly **200** includes a housing **202**, one or more batteries or other energy supply **221**, one or more capacitors or other energy storage **222**, a microprocessor **223**, a coil assembly **224** (which functions as a motor stator), and one or more Hall effect sensors **225**. Exemplary disposable assembly **300** includes a baseplate **350** supporting components such as a magnetic motor rotor **331**, a gear train **332** including lead screw drive gear **333** in a reservoir support block **337**, and a lead screw **334** attached to plunger **335**, which is positioned in a medicament reservoir **336** that is mounted to the reservoir support block **337**. The exemplary plunger **335** includes a core and a plurality of seals on the core. A cover **302**, under which some or all of the magnetic motor rotor **331**, gear train **332** (with drive gear **333**), lead screw **334**, plunger **335**, and medicament reservoir **336** are located in various embodiments, may be mounted to the baseplate **350**.

(59) The lead screw drive gear **333**, lead screw **334**, plunger **335**, medicament reservoir **336** and reservoir support block **337** may also be referred to collectively as a “reservoir assembly.” Other exemplary reservoir assemblies, durable assemblies, and seal assemblies that may be employed in, for example, infusion system **100** are described below with reference to FIGS. **9D-18C**.

(60) The exemplary disposable assembly **300** may be secured to the exemplary durable assembly **200**, as shown in FIGS. **1A** and **2**. To that end, the exemplary housing **202** includes a top wall **204**, bottom walls **206a** and **206b** and a side wall **208** that together define a relatively thin housing portion **210** and a relatively thick housing portion **212**. An indentation **214** is formed in the relatively thick portion **212**. The exemplary cover **302** includes top walls **304a** and **304b** and a side wall **306** that together define a relatively thin cover portion **308** and a relatively thick cover portion **310**. A portion of the baseplate **350** is not covered by the cover **302**, thereby defining a recess **312** that is bordered by a wall **314** that extends around the baseplate (see also FIG. **4B**). When the durable and disposable assemblies **200** and **300** are secured to one another in the manner illustrated in FIG. **1A**, the relatively thick portion **212** of the housing **202** will reside in the recess **312** of the disposable assembly **300** (with the wall **314** in the indentation **214**). The relatively thin portion **210** of the housing **202** will reside on the top wall **304b** of the cover **302**. The cover **302** also includes a projection **316** that mates with a recess **216** on the housing **202**. Additionally, as is discussed in greater detail below, the disposable assembly **300** may be configured for different medicaments, such as different medicament concentrations, different medicament amounts, or different modes of system operation.

(61) In other implementations, the cover **302** may be configured to cover fewer than all of the components on the baseplate **350**. For example, a cover may be configured such that the magnetic motor rotor **331** and a portion of the gear train **332** are not under the cover, while the remaining components are under the cover. In still other implementations, the cover **302** may be omitted and the durable assembly **200** may be configured to cover all of the components on the baseplate **350**. In yet other implementations, what is referred to in the present application as the “durable” assembly, may be disposable, resulting in a fully disposable system.

(62) As discussed in U.S. patent publication number 2012/0078170 described above, and in U.S. application Ser. No. 13/300,574, filed Nov. 19, 2011, and corresponding U.S. patent publication

number 2012/0184907, and in U.S. application Ser. No. 13/475,843, filed May 18, 2012, and corresponding U.S. patent publication number 2013/0138078, each of which are incorporated by reference in their entireties, ambulatory infusion systems that employ a reservoir on a baseplate may be configured for different types of use. For example, disposable assembly **300** may be adhered to the patient's skin and may be used in conjunction with a cannula (not shown) that is operatively connected to the reservoir **336** so that the system **100** may be deployed as a "patch-pump," as shown in FIG. 2A. Alternatively, as shown in FIGS. 2B, the baseplate **350** of disposable assembly **300** may be configured to operably connect the reservoir **336** to an infusion set **382** (e.g., by way of the illustrated infusion set tube and a connector **380** shown in FIGS. 1B and 2) so that the system **100** may be deployed as a "pocket pump," a "belt-worn pump" or some other wearable pump. In other words, using the same durable assembly **200**, the user may configure the system for use as "pocket pump" or a "patch pump" by simply selecting the appropriate disposable assembly and attaching the disposable assembly to the durable assembly. The user may also switch from one configuration to another, by simply removing one disposable assembly and replacing it with another disposable assembly. The connector **380** may also be used as a fill port, as discussed below. (63) It should therefore be noted that the present inventions include kits that contain various combinations of disposable assemblies, where at least two of the disposable assemblies may be different. Additionally or alternatively, kits or other packages may include various disposable assembly components, such as an infusion set and/or cannula inserter. Kits may also include a durable assembly. The disposable assemblies in such kits may also include the detection/identification instrumentalities discussed below. The components of the present kits (e.g., combination of various disposable assemblies and/or components) may be stored in a common package, with individual packages for each component if necessary, and provided to the user in the common package. Other components that may be provided in such kits include, but are not limited to, inserters that are preloaded with a cannula, and cleaning swabs. A recharger may also be provided in a kit that includes a durable assembly.

(64) In addition to disposable assembly packaging and labeling, the different disposable assemblies may include visual cues to differentiate the various disposable assemblies. For instance, disposable assemblies with different concentrations of medicament or different medicament fill volumes may use different colors for the reservoir and/or baseplate of the disposable assembly, or mechanical features that ensure disposables are only able to attach to correctly programmed durables.

(65) It should also be noted here that, but for the issue of priming, the dispensing procedures associated with an infusion system "patch pump" configuration, which may include a durable assembly **200** and a disposable assembly **300**, are substantially the same as the dispensing procedures associated with a "pocket pump" configuration, which may also include an infusion set **382** (see FIGS. 2B). With a "patch pump" configuration, priming is not necessary because the volume of the associated cannula will be very small and there is a direct connection between the cannula and the medicament reservoir. Priming is, however, required to fill the infusion set tube (FIG. 2B) in a "pocket pump" configuration prior to the onset of medicament delivery. For instance, 20-30 μ l may be required to fill the entire infusion set tube and, accordingly, the priming procedure may involve the rapid delivery of 10-15 IUs of U-500 insulin to the tube. The present inventors have determined that it would be advantageous to prevent users from initiating a priming procedure when the system is in the "patch pump" configuration, with a cannula positioned to deliver medicament essentially directly from the medicament reservoir to the patient, because rapidly delivering 10-15 IUs of insulin to the patient could adversely affect patient health.

(66) To prevent such undesirable outcomes, and for user convenience in other situations involving the choice between a variety of disposable assemblies (such as disposable assemblies with reservoirs containing different medicaments, different concentrations of a medicament, and/or varying amounts of medicaments), at least some of the present disposable assemblies may be provided with a baseplate identification device and at least some of the present disposable

assemblies may be provided with structure that cooperate with a baseplate identification device in such a manner that the durable assembly microprocessor/controller can make a “baseplate type” determination. Exemplary baseplate identification instrumentalities and methodologies may be as described in aforementioned U.S. patent publication numbers 2012/0078170, 2012/0184907, and 2013/0138078. In addition, baseplate identification may be performed mechanically. For instance, a pin or rib may prevent attachment of certain disposable assemblies with certain durable assemblies. Additionally or alternatively, certain durable assemblies will simply not function with certain disposable assemblies.

(67) Alternatively, the patient or a clinician may program the system, such as via a remote control, to indicate the type of disposable assembly attached. In a manner such as this, a patient can access a variety of medicaments for use with a single durable assembly.

(68) Once the “baseplate type” determination is made (e.g., “patch pump” disposable assembly **300** versus a “pocket pump” with infusion set **382** attached), the durable assembly will proceed in a manner, or mode of operation, that is appropriate for the attached disposable assembly. For example, if “patch pump” disposable assembly **300** is detected, the durable assembly controller will not include priming as part of the delivery process and, in some implementations, will prevent the user from manually implementing a priming procedure. If, on the other hand, a “pocket pump” disposable assembly is detected, then the delivery process may include appropriate priming of the infusion set tube.

(69) Whether configured as a “pocket pump” or a “patch pump,” the system may be configured to provide basal delivery of medicament in accordance with a delivery profile provided by a physician by way of a clinician's programming unit. For example, the system may include a program that stores a number of delivery profiles (e.g., delivery profiles associated with a 24-hour delivery cycle, delivery profiles for particular situations such as sleep or illness, and the like). Each delivery profile specifies multiple doses (or pump “operations”) over time, e.g., a particular number of doses at particular times or a particular number of doses per unit time. In some implementations, a dose may be the volume associated with the minimum controllable displacement of the plunger **335**. The system may also be configured to provide bolus delivery in response to an instruction from a patient remote control **1000** (FIG. 2A). A bolus instruction may come in response to a high glucose level measurement in the case of a diabetic patient, an increase in pain level in the case of a pain management patient, or some other symptom. The system may also be configured to perform other functions, such as ending medicament delivery in response to instructions from patient remote control **1000**.

(70) The present infusion pumps may be used in conjunction with a wide variety of remote controls. Such remote controls may be used to, for example, allow the user to transmit instructions to the durable assembly **200** or facilitate communication between durable assembly **200** and the user (e.g., an alarm condition message or other message concerning the conditions of system **100**). An exemplary remote control **1000** (FIG. 2A) may be configured to facilitate one, some, or all of the following operations: (1) turning the remote control **1000** on or off, (2) associating (or “assigning”) the remote control **1000** to the durable assembly **20**, (3) obtaining status information such as medicament level, battery charge level, and/or alarm conditions, (4) silencing the durable assembly alarm, (5) selecting options that may be associated with the durable assembly alarm such as type of alarm (audible, palpable, and/or visible) and strength/volume of alarm, (6) connecting remote control **1000** to a computer to, for example, update remote control or durable assembly firmware, load and delete delivery profiles stored in the durable assembly or remote control, and otherwise reprogram the durable assembly or remote control, (7) selecting medicament options such as medicament concentrations, (8) selecting and initiating a stored medicament delivery profile, (9) increasing and decreasing medicament dose rate, and/or (10) pausing a dispensing operation. A user may pause delivery in order to remove or replace a patient applied structure (e.g., a disposable assembly), adjust for a current or anticipated changed body condition (e.g., low

glucose, vigorous exercise), follow a physician's suggestion, or disconnect the durable assembly from the body for any other reason.

(71) The exemplary remote control **1000** (FIG. 2A) may be configured to generate an indicator, based on information from a microprocessor **223** for durable assembly **200**, that is indicative of, for instance, the amount of time remaining in the current dispensing program, the amount of time until the next disposable assembly replacement, etc. The indicator may be audible, visible, palpable, or combinations thereof. A time remaining indicator may be useful for a variety of reasons. For example, knowledge of the time remaining prior to next disposable assembly replacement allows the patient to determine, based at least in part on the current time of day and upcoming events (e.g., travel or sleep), whether or not it would be more convenient to replace the disposable assembly at a time prior to the end of the dispensing program.

(72) As described above, parts of the present systems may be considered the reusable parts, while other parts may be considered the disposable parts. In the illustrated embodiments, the durable assembly **200**, which may include structures such as microprocessor **223** and coil assembly **224**, is reusable, while exemplary disposable assemblies **300**, which may include structures such as a motor rotor **331** and reservoir **336** on a baseplate **350**, are disposable. In other embodiments, the present systems may be fully disposable.

(73) With respect to dimensions, some embodiments of the exemplary infusion pump system **100** may have the following dimensions: length dimensions of 35 mm $\pm$ 1.0 mm, 35 mm $\pm$ 0.10 mm, or 35 mm $\pm$ 5.0 mm; width dimensions of 30 mm $\pm$ 1.0 mm, 30 mm $\pm$ 0.10 mm, or 30 mm $\pm$ 5 mm; and overall thickness or height dimensions of 8.5 mm $\pm$ 1.0 mm, 8.5 mm $\pm$ 2 mm, or 8.5 mm $\pm$ 0.10 mm. Suitable housing materials include, but are not limited to, plastic or other materials having a modulus of elasticity of 0.2-1.0 million psi.

(74) Exemplary durable assembly microprocessors and associated circuitry; rechargeable batteries and associated battery rechargers and recharging methods; battery and recharging management; temperature sensors; and exemplary alarms and alarm conditions are described in more detail in aforementioned U.S. patent publication numbers 2012/0078170, 2012/0184907, and 2013/0138078.

(75) Turning now to FIGS. 3A and 3B, an exemplary durable assembly **200** may include a power source such as one or more batteries **221**, temporary power storage such as one or more capacitors **222** (see FIGS. 2 and 5B), a controller such as microprocessor **223**, a coil assembly **224**, and a hall effect sensor **225**. Those of skill in the art will appreciate that including the motor's coil assembly **224** and all other electronics within the durable assembly **200** reduces the cost and complexity of disposable assembly **300**. In addition, the microprocessor **223** provides flexibility to include features such as user data storage, programs, programmability, adjustability, a display, buttons, wireless communication protocols, or the like to the pump **100**. Durable assembly **200** may also be molded with locking features that snap onto the disposable assembly **300**, but that also allow removal of the durable assembly **200** from the disposable assembly **300** either while the disposable assembly remains in place on the patient (after medicament delivery has been paused), or after the entire system has been removed from the patient.

(76) The power source may be one or more commercially available batteries, such as a commercially available zinc-air battery or lithium polymer battery. The batteries may be selected to have sufficient capacity to operate the system for certain delivery amounts or delivery times, such as for over 400 units of delivered insulin. The optional power storage may be one or more commercially available capacitors or super-capacitors or other temporary storage device(s).

(77) Turning now to FIGS. 4A and 4B, an exemplary disposable assembly **300** may include baseplate **350** and components such as a reservoir **336**, a plunger **335** within the reservoir and connected to lead screw **334**, and a magnetic motor rotor **331** mechanically attached through gear train **332** to affect rotation of the lead screw drive gear **333**, which causes translation of the lead screw **334** and plunger **335** within reservoir **336**. The cover **302** is positioned over these

components in the illustrated embodiment. The exemplary baseplate **350** includes an adhesive backing for attachment to the patient with a removable adhesive cover. The baseplate **350** may also be molded with baseplate locking features that snap onto the durable assembly **200** (such as magnets molded into the housings of each assembly), and that also allows removal of the durable assembly **200** from the disposable assembly **300**.

(78) Referring to FIGS. **2** and **4B**, the exemplary reservoir **336** includes a barrel **338** with an inner surface **340** defining a fluid storage volume **342** and an oval cross-section, but other shapes (such as circular) are possible as is discussed below with reference to FIGS. **10A-18C**. A plunger **335** with a matching cross-sectional shape fits within the barrel and carries a fluid seal such as, but not limited to, o-rings, to seal the medicament within the storage volume **342**. The exemplary plunger **335** is formed from rubber and includes three o-ring seals. The reservoir **336** includes the aforementioned connector **380** that may be used for filling reservoir **336**, or for attaching a cannula for “patch-pump” type configurations, or for connecting (potentially via an appropriate adapter(s)) an infusion set for “pocket-pump” type configurations. The plunger **335** moves within the barrel **338** to vary the volume of medicament within the storage volume **342**. Reservoir **336** may be, for instance, prefilled (or user-filled) with U-500 insulin in various volumes to suit the patient use profile. In other instances, lower concentrations of insulin, such as U-100 insulin and U-200 insulin, may be employed. A plug may be inserted in the connector **380** to maintain a sterile environment until use. The patient would remove the plug prior to use, in those instances.

(79) Additional exemplary baseplates for use with the disposable assemblies of the present inventions, as well as exemplary cannula designs, fluidic connection between a medicament reservoir and the cannula, cooperation between the cannula and disposable assemblies (for instance, to prevent axial movement of the cannula relative to the baseplate and patient), attachment of an infusion set to the reservoir of the disposable assembly, configurations and uses of a non-delivery baseplate, arrangements and structures for attaching disposable and durable assemblies, skin adhesive designs, and various occlusion sensors, may be as described in U.S. patent application Ser. No. 12/890,207, filed Sep. 24, 2010 and corresponding U.S. patent publication number 2012/0078170, as well as aforementioned U.S. patent publication numbers 2012/0184907 and 2013/0138078.

(80) Turning now to FIGS. **5A-5C** and the illustrated two-piece motor, the motor's coil assembly **224** (and a Hall effect sensor **225**) of the durable assembly **200** are positioned above the magnetic motor rotor **331** that is part of the disposable assembly **300**. An exemplary multi-pole motor rotor **331** may be disc-shaped and have a 9.8 mm outer diameter, 5.2 mm inner diameter, and 0.8 mm thickness. Another example motor rotor may have an 11 mm outer diameter, 5 mm inner diameter, and 1.2 mm thickness. Multi-pole motor rotors of this type typically cost less than 5 cents per piece, helping control the total cost of disposable assembly **200**. The motor rotor **331** is also parallel to the baseplate **350**, i.e., the motor rotor axis of rotation is perpendicular to the baseplate, in the illustrated embodiment. The microprocessor **223** directs rotation of motor rotor **331** by sequentially energizing the coils of motor coil assembly **224** to create an electromagnetic torque coupling between the motor coil assembly **224** and the motor rotor **331**. The position/orientation of the rotor's poles relative to the rotating magnetic field generator (coil assembly **224**) is measured by back EMF, a rotary encoder, a hall effect sensor **225** (FIG. **5A**), or the like. For instance, a Hall effect sensor mounted on the coil windings may be used to supply the microprocessor a count, a tachometer signal, or rotor position, allowing low-cost closed-loop control of the rotor speed. Brushless motors of this type are typically 85-90% or more efficient, and run very cool. While there may be variations in construction, the face-to-face stator coils and flat rotor plate shown in FIGS. **5A-5C** provide a compact design. In addition, more coils and/or Hall effect sensors may be used.

(81) As can best be seen in FIG. **6**, between the motor coil assembly **224** and motor rotor **331** is a gap **240**. Some or all of the gap **240** may be defined by (and occupied by) portions of the housing

202 and the cover **302**, i.e., the housing bottom wall **206a** and the cover top wall **304b** in the illustrated implementation. In other implementations, the gap **240** between the between the motor coil assembly **224** and motor rotor **331** may be occupied by only a portion of the durable assembly housing, or only a portion of the disposable assembly cover, or no structure at all and may simply be an air gap. The size of the gap, which is defined by the distance between the motor coil assembly **224** and the motor rotor **331**, is typically about 0.5 mm to 2.0 mm. As such, there is no gear engagement or other mechanical connection between the durable assembly **200** and disposable assembly **300**. And as described earlier, all electronics may be positioned within the durable assembly **200**, with the energy needed by the disposable assembly **300** transferred by electromagnetic torque coupling, which is a coupling without direct mechanical coupling or electrical contact from the durable assembly **200**. This exemplary design affords the additional advantage of being relatively simple to make waterproof, or at least water resistant.

(82) As described above, rotation of motor rotor **331** drives gear train **332**, causing rotation of lead screw drive gear **333**, which in turn affects translation of the lead screw **334** and plunger **335**, which is attached to lead screw **334**. In this manner, electromagnetically generated torque is created when electromagnetic energy supplied by durable assembly **200** is transformed into mechanical forces within the disposable assembly **300** that advance plunger **335**. A ratchet (not shown) or other similar device may be used to prevent back drive of gear train **332**. As plunger **335** is driven through reservoir **336**, medicament is dispensed precisely, corresponding to the precision movements of the gears and motor rotor. With the entire gear train, lead screw drive gear, lead screw, and plunger all permanently contained in the disposable assembly **300**, there is no need to retract any plunger components into the durable assembly **200** prior to separation from the disposable assembly **300**. As a result, a further advantage of this exemplary design is greatly reduced energy consumption, which allows use of, for instance, a primary battery(ies) as a power source.

(83) Use of an exemplary system **100** will now be described. At the most basic level, a patient's use of the exemplary infusion pump systems (e.g., system **100** in FIGS. **1A-2B**) involves obtaining a new disposable assembly **300**, connecting the disposable assembly to the durable assembly **200**, peeling the liner from the baseplate adhesive layer, gaining subcutaneous access, and initiating a medicament delivery operation. In some instances, use may involve additional steps such as attaching a cannula to connector **380** of the disposable assembly and removing a cannula cap, if necessary. Various aspects of the basic operation of the present systems are described below. Operation of a system does not necessarily require all of the steps each time the system is deployed, and the order of some of the steps may be changed. Operation is also discussed below, in the exemplary context of the above-described durable assembly **200** and disposable assembly **300** used as a patch pump, through the use of a flow chart (FIG. **8**). The discussion is, however, equally applicable to other patch pump implementations, as well as to pocket pump implementations with minor variations. Also, unless otherwise indicated, the actions and determinations performed by the durable assembly **200** are controlled by the durable assembly microprocessor and further references to the controller are limited in the interest of brevity.

(84) Referring to FIG. **8**, use of the present systems may involve removal of a disposable assembly from a durable assembly and the replacement of the disposable assembly. This may occur when the medicament reservoir is empty (as described in more detail in U.S. patent application Ser. No. 12/890,207 and corresponding U.S. patent publication number 2012/0078170) (Step **S101**) and a “replace disposable assembly” message or alert is presented (Step **S102**), or when the durable assembly controller receives a user-initiated “replace disposable assembly” signal from a remote control **1000** (Step **S103**). The user may desire to replace a disposable assembly before the medicament reservoir is empty for a variety of reasons such as, for example, to accommodate the user's sleep or travel schedule, when the medicament exhibits a loss of effectiveness, when a dispensing problem arises, or due to a prescribed change in medicament.

(85) The user may then obtain, possibly from storage in a refrigerator depending on medicament requirements, a new pre-filled disposable assembly **300** or may then obtain a new disposable assembly and fill the disposable assembly with medicament (Step **S104**). The durable assembly **200** and disposable assembly **300** may then be removed from the skin, separated, and the disposable assembly **300** discarded (Steps **S106** and **S107**).

(86) Next, the new disposable assembly **300** may be attached to the durable assembly **200** (Step **S109**). The user should clean the skin surface **S** onto which the baseplate **350** of disposable assembly **300** will be adhered (FIG. 7, and Step **S116** of FIG. 8). Then the user peels off the baseplate adhesive liner to expose the baseplate adhesive layer (Step **S117**) and removes cannula cap (when present) (Step **S118**). In the exemplary use of FIG. 8, the disposable assembly **30** is supplied with a cannula in fluid communication with the reservoir storage volume. In other embodiments, a cannula inserter may be attached to the system, which may be triggered to insert the cannula after the system is placed against the skin. Exemplary inserters are described in U.S. patent publication number 2013/0138078.

(87) Returning to the steps in FIG. 8, the system **100** including durable assembly **200** and disposable assembly **300** may be positioned over a suitable body location and pressed gently to adhere the adhesive layer to the skin surface **S** and, once the system has been adhered (Step **S119**), the inserter may be actuated to position the end of a cannula below the skin. It should be noted that in those implementations which do not include an inserter, and instead simply include a hollow needle (or a cannula and removable trocar arrangement) that projects outwardly from the bottom surface of the system, the user need only adhere the adhesive layer to position the needle or cannula below the skin. The trocar, if employed, may then be removed. Finally, if necessary, the remote control **1000** may be used to initiate a particular medicament delivery operation (Step **S120**). The delivery operation may follow a predetermined delivery profile (e.g. a particular basal rate, a series of time-spaced bolus deliveries, or some combination thereof) that is equated to motor rotor rotations, at particular rates and times, required to deliver medicament in accordance with the profile. Alternatively, the profile may be input by the user with the remote control **1000** and stored by the durable assembly microprocessor. For example, the remote control may store a number of different delivery profiles and bolus deliveries from which the patient can choose. Such profiles may correspond to, for example and depending on the medicament, days where vigorous exercise is expected, days where it is not, incidences of increased pain, etc. Alternatively, or in addition, the profile stored in the durable assembly microprocessor may be set by a clinician's programming unit. In such a case, as in the case of different disposable assemblies **300** provided with different specified delivery rates, a remote control may not be needed to initiate, e.g., basal delivery.

(88) The discussion above is also applicable to use of the "pocket pump" system as shown in FIG. 2B. Minor variations in the above-described procedure include, for example, use of an infusion set **382** instead of a cannula, attaching the infusion set to connector **380**, potentially via an adapter (which may vary with the type of infusion set **382**), and priming of the infusion set tube.

(89) Another exemplary ambulatory infusion system, which is generally represented by reference numeral **100a** in FIG. 9A, includes a durable assembly **200a** and a disposable assembly **400**. System **100a** is substantially similar to system **100**. Here, however, the intersection of the top walls is primarily linear. Additionally, the disposable assembly **400** has a recess **316a** which mates with a corresponding projection **216a** on the durable assembly **200a**. The projection **216a** and recess **316a** are located at the outer perimeter of the assembled system **100a**.

(90) Exemplary durable assembly **200a**, shown in more detail in FIGS. 9B and 9C, may include a housing **202a**, one or more batteries or other energy supply **221a**, one or more capacitors or other energy storage (not shown), a microprocessor (not shown), and a coil assembly **224a** including one or more Hall effect sensors (not shown). Exemplary disposable assembly **400**, shown in more detail in FIGS. 9D and 9E, may include a baseplate **450** supporting components such as a magnetic motor rotor **431**, a gear train **432** including lead screw drive gear **433**, and a lead screw **434** attached to

plunger assembly **435** which is positioned in a medicament reservoir **436**. The magnetic motor rotor **431** may be mechanically attached through gear train **432** to affect rotation of the lead screw drive gear **433**, which causes translation of the lead screw **434** and the plunger **435** within reservoir **436**. Reservoir **436** may be, for instance, prefilled with U-500 insulin or U-100 insulin or other concentrations of insulin to suit different patient use profiles, or may be user-fillable by way of a fill port **415**. A reservoir outlet **438** is in fluid communication with reservoir **436**. A septum **446**, which may form part of a trocar seal assembly (such as those described below), is positioned within the reservoir outlet. Disposable assembly **400** may be secured to durable assembly **200a**, as shown in FIG. **9A** and as further described in U.S. provisional patent application Ser. No. 62/057,273, corresponding U.S. patent application Ser. No. 14/869,906, and corresponding U.S. patent publication number 2016/0089491.

(91) The reservoirs may be, but are not required to be, prefilled. Prefilled reservoirs are advantageous for a variety of reasons. By way of example, but not limitation, some users prefer to avoid reservoir filling procedures because they are inconvenient and tend to involve needles. User-based refilling also increases the likelihood that air bubbles will be introduced into the reservoir, while prefilling by the manufacturer of the reservoir and/or the medicament can be accomplished without any substantial introduction of air bubbles using, for example, a vacuum filling procedure. Nevertheless, user-filled reservoirs may be employed in some instances. A variety of exemplary medicament reservoirs, including those that include pressure sensors (such as for sensing occlusion) and other sensors, are described in more detail in aforementioned U.S. patent publication numbers 2012/0078170, 2012/0184907, and 2013/0138078.

(92) While a prefilled reservoir would greatly improve the ease of use of patch and pocket pump technology, there are several challenges to providing such prefilled reservoirs. By way of example but not limitation, long-term storage of insulin has traditionally used glass containers with bromobutyl rubber stoppers, and this has been applied to prefilled insulin pens using glass syringe barrels with bromobutyl plungers. The high coefficient of friction of bromobutyl on glass requires a coating of silicone oil on the interior of the reservoir, so the plunger may slide easily in the barrel during dispensing. Alternatively or additionally, as described in more detail in U.S. provisional patent application 62/117,565 filed Feb. 18, 2015, corresponding U.S. patent application Ser. No. 15/042,093, filed Feb. 11, 2016, and corresponding U.S. patent publication number 2016/0235913, the reservoir assembly may utilize a dual seal system having a static seal that minimizes water vapor loss during storage or pre-dispensing, and a dynamic seal that provides a low glide/break force when the plunger is moving during dispensing.

(93) As another non-limiting example, any opening used to introduce medicament while filling the reservoir must then be properly sealed to prevent pre-dispensing fluid leakage and to limit water vapor loss during any storage period. As yet another non-limiting example, the filled reservoir also needs proper outlet seals along the dispensing path, designed for medicament compatibility, and designed to limit vapor loss during any storage of a prefilled reservoir; and those seals need a passage, valve(s) or other means to ultimately allow medicament dispensing. For instance, and as seen generally in FIGS. **10A** and **10B**, an exemplary trocar seal assembly **540** may be positioned in a reservoir outlet **538** that is in fluid communication with reservoir **536**. The trocar seal assembly **540** must prevent pre-dispensing fluid loss once the reservoir **536** is user-filled, or if pre-filled before being supplied to the user, must prevent vapor loss during storage, but in both cases, allow dispensing of medicament from reservoir **536** through the reservoir outlet **538** and trocar seal assembly **540** once properly activated by a patient. The exemplary trocar seal assembly **540** includes a cannula **541**, a cannula insert **542**, a combined drum/septum **543a** (with a drum **543** and integral septum **546**), and a trocar (not shown), and functions in the manner described below with reference to FIGS. **11A-11F**. Trocar seal assembly **540** (and other trocar seal assemblies described below) may be incorporated into an infusion pump in, for example, the manner illustrated in FIG. **9E**.

(94) A more specific exemplary embodiment is shown in FIGS. 11A-11F, where reservoir outlet **638** with an inner surface **660** is in fluid communication with a fluid reservoir (such as, for example, the reservoir **536** in FIG. 10A or the reservoir **636** in FIG. 12) via reservoir connection **639** (or similar, such as reservoir connection **839** best seen in FIGS. 14B and 14C). A drum **643**, made of rubber or similar conformable material, and which includes an outer surface **662**, an inner surface that defines a drum outlet lumen (or “drum outlet”) **664**, drum seals **644** on the outer surface, and trocar seals **645** on the inner surface, is positioned within reservoir outlet **638**. Drum **643** also includes a drum inlet lumen (or “drum inlet”) **649** in fluid communication with reservoir connection **639** and drum outlet **664**. The trocar seal assembly **640** (FIGS. 11E and 11F) comprises the drum **643**, a septum **646** positioned above drum **643**, a cannula **641** and cannula insert **642**, and a trocar **647** positioned through septum **646**, drum **643**, cannula insert **642**, and cannula **641**. Trocar **647** includes an elongate rod, with an outer surface **666**, and a sharp end **668**. As with drum **643**, septum **646** may also be made of rubber, such as silicone rubber, or the like. Cannula **641** may be made of polytetrafluoroethylene (PTFE), such as TEFLON®PTFE, or other biocompatible polymeric material, and cannula insert **642** may be made of metal, such as stainless steel, or other relatively rigid biocompatible material. Cannula insert **642** holds cannula **641** in place and seals cannula **641** against reservoir outlet **638**. Additionally, cannula insert **642** guides trocar **647** into place within cannula **641** during assembly and prevents the sharp end of the trocar from piercing the cannula. The fluid reservoir may be made of cyclic olefin polymer (COP) or other biocompatible polymeric material and reservoir outlet **638** may be made of cyclic olefin polymer (COP) or other biocompatible polymeric material.

(95) In another exemplary embodiment, shown in FIG. 12, the drum and septum are integrated into a combined drum/septum **643a** with drum **643'** and integral septum **646'**. The entire body **643a** may be made of rubber or other conformable material, or a technique such as over-molding may be used to form body **643a** of multiple materials. The other components shown in FIG. 11A-11F may be used interchangeably with this exemplary embodiment.

(96) In the above exemplary embodiments, if the reservoir is filled with medicament before delivery to a user (rather than user-filled as described elsewhere in this application) trocar **647** is the primary packaging seal for the medicament, so the medicament cannot travel from the fluid reservoir through the reservoir outlet **638** to cannula **641** by way of, in the illustrated example, reservoir connection **639**, drum inlet **649** and drum outlet **664**. In the case of user-fill, the trocar **647** acts as a medicament seal and similarly prevents medicament leakage until the user activates the pump. In either case, drum seals **644** engage outlet inner surface **660** to seal the interface between the reservoir outlet **638** and drum **643** or (combined drum/septum **643a**). While trocar **647** is in place in a first (or “seal”) position (FIGS. 11E and 11F), trocar seals **645** engage trocar outer surface **666** to seal the interface between trocar **647** and drum **643** (or combined drum/septum **643a**), thereby preventing fluid flow through drum outlet **664** and to cannula **641**. Trocar **647** may be made of stainless steel, rigid plastic, ceramic, or similar rigid biocompatible material, and is used to penetrate the skin and a short distance into the flesh, to make a channel for cannula **641**. In at least some instances, the associated pump will be pressed against the patient's skin, thereby driving cannula **641** and trocar **647** below the skin surface. Once cannula **641** is in place, trocar **647** may be moved to a second (or “unseal”) position to permit fluid flow from reservoir outlet **638** to cannula **641**. For example, trocar **647** may be retracted into the pump **100/100a**, or removed from pump **100/100a** and the remainder of the trocar seal assembly **640** (FIG. 11D) through septum **646** (or combined drum/septum **643a**), thereby removing the trocar from drum outlet **664** to permit fluid flow. Septum **646** (or the septum portion of combined drum/septum **643a**) will then seal end of drum outlet **664** opposite the cannula **641**.

(97) Another exemplary embodiment of a trocar being used as a medicament seal or a primary packaging seal is shown in FIGS. 13A-13Q. This exemplary trocar seal assembly **740** includes a drum **743**, a cannula **741**, and a trocar **747**. Drum **743** includes an outer surface **762**, drum seals

744 on the outer surface, an inner surface that defines a drum outlet lumen (or “drum outlet”) **764**, and cannula seals **745** on the inner surface (FIGS. **13B** and **13C**). Drum **743** is positioned within reservoir outlet **738**, and further includes a drum inlet **749** in fluid communication with reservoir connection **739** and drum outlet **764**. Cannula **741** and trocar **747**, when in the pre-dispensing or storage position, are positioned within drum **743** to seal the fluid path through drum **743** that is defined by drum outlet **764**, as described in more detail below.

(98) Trocar **747** includes an elongate rod, with an outer surface **766**, and a sharp end **768**, may be made of stainless steel, rigid plastic, ceramic, or similar rigid biocompatible material, and may be attached to button **700**, as is best seen in FIGS. **13F-13G**. For instance, trocar **747**, which may have an outer surface **766** with a diameter of about 0.20 mm (about 0.008 inch), may be adhered to button **700**, which may have an outer diameter of about 10 mm (about 0.4 inch). As another example, trocar **747** may be insertion molded into button **700**, which may be made of COP, Delrin, polycarbonate, or other injection moldable polymeric material. As best seen in FIGS. **13D**, **13E**, **13K**, **13M**, and **13N**, trocar **747** carries cannula **741**, which may have an outer surface **770** that is in contact with the cannula seals **745** and has a diameter of about 0.50 mm (0.020 inch). Cannula **741** may be made of peek, TEFLON®PTFE, or other polymeric material. Button **700** is attached to stem **702**, which carries a stem hook **704**, such as a tab or rim. As may be best appreciated from the section views of FIGS. **13G**, **13H**, and **13J**, all these pieces may be, but are not necessarily, made of the same material and may even be made as one piece, such as by injection molding all at once. Stem **702** slides within tube **720** (FIGS. **13P-13Q**), which may be made of the same or similar materials as button **700** and stem **702**. Tube **720** has a stem catch **722** that latches with stem hook **704** to prevent stem **702** from disengaging from tube **720**. Similarly, tube **720** slides within cavity **755** of reservoir outlet **738** (FIG. **13A**), which has a tube catch **742** that latches with tube hook **724** to prevent tube **720** from disengaging from reservoir outlet **738**.

(99) When the associated disposable assembly is filled with medicament but the pump is not yet dispensing, trocar seal assembly **740** acts as a pre-dispensing seal or a primary packaging seal, with button **700** and tube **720** at their highest, fully extended positions, as depicted in FIG. **13A**. In this storage or pre-dispensing configuration, the end of cannula **741** and tip of trocar **747** are positioned above the bottom of reservoir outlet **738**, as seen in FIG. **13A**. For instance, the tip of trocar **747** and the end of cannula **741** may be about 0.1 mm above the bottom of reservoir outlet **738**. In this storage or pre-dispensing configuration, where cannula **741** and trocar **747** are in first (or “seal”) positions, cannula inlet **758** is not aligned with drum inlet **749**, so medicament cannot flow through cannula inlet **758** and into cannula passage **759**. In addition, cannula seals **745** engage cannula outer surface **770**, thereby preventing flow out between the inner surface of drum **743** and the cannula outer surface. Cannula passage **759** (FIGS. **13M** and **13N**) is a channel along the inside of cannula **741** that allows passage of medicament down the length of cannula **741** and trocar **747**, and out the distal end of cannula **741**.

(100) To initiate the flow of medicament through the pump, button **700** is pressed until stem **702** is entirely within tube **720**, tube **720** is entirely within reservoir outlet **738**, and button **700** is seated in the top, wide portion of cavity **755** within reservoir outlet **738**. Pressing down button **700** causes button spring **706** to compress about 0.9 mm (0.035 inch) (see FIGS. **13D** and **13E**, and also **13H** and **13J**), which in turn causes the distal tip of trocar **747** to extend about 0.9 mm (0.035 inch) past the distal end of cannula **741** (see FIGS. **13D** and **13E**). When button **700** is pressed all the way down into the top, wide portion of cavity **755** to drive cannula **741** and trocar **747** to the second (or “unseal”) position, the distal end of trocar **747** penetrates the skin and extends about 6-8 mm past the bottom of reservoir outlet **738**, and into the patient's flesh, carrying the distal end of cannula **741** with it. In this position, cannula inlet **758** is located between the cannula seals **745** above and below drum inlet **749** and is substantially aligned with drum inlet **749**, which allows medicament to flow through reservoir connection **739**, through drum inlet **749**, through cannula inlet **758**, into cannula passage **759**, and out cannula **741** (see FIGS. **13K**, **13M**, and **13N**). Cannula passage **759**

may be a channel with a width W of about 0.15 mm (about 0.006 inch) and a depth D of about 0.05 mm (about 0.002 inch). Cannula passage **759** is closed at the proximal end so medicament cannot flow up cannula **741** into button **700** (see FIG. 13M). When button **700** is released by the user, button spring **706** will drive the button (and trocar **474**) as short distance (e.g., 0.9 mm) away from outlet **738**. The trocar **747** will move towards, but not to, the first position, while the cannula **741** will remain in second position.

(101) Another exemplary embodiment of a trocar being used as a medicament seal or a primary packaging seal is shown in FIGS. 14A-14D. This exemplary trocar seal assembly **840** includes a drum **843**, a cannula **841**, cannula insert **842**, and a trocar **847**. Drum **843** includes an inner surface that defines a drum outlet lumen (or “drum outlet”) **864**. A septum **846** may be positioned above drum **843**, or the drum and septum may be integrated into a combined drum/septum (not shown). The exemplary trocar **847** includes an elongate rod, with an outer surface **866**, and a sharp end **668**. In this exemplary embodiment, reservoir outlet **838** is created as part of the baseplate **850** of disposable assembly **800** (FIGS. 14A-14B), reservoir connection **839** is a nipple protruding from and may be created as part of the reservoir **836** (FIGS. 14B-14C), and drum **843** includes a retainer **849a** that holds drum **843** onto reservoir connection **839** and creates a sealed fluid connection between the nipple of reservoir connection **839** and the drum inlet **849** (FIGS. 14B-14D). Drum **843** is positioned within reservoir outlet **838** and includes trocar seals **845** (FIG. 14B) that engage the outer surface **866** of trocar **847** to form a seal around trocar **847** as described in the embodiments above. Reservoir **836** and reservoir connection **839** may be made of COP or other biocompatible polymeric material, drum **843** or the previously described optional combined drum/septum may be made of bromobutyl rubber or similar elastomeric material, and septum **846** may be made of silicone rubber or other resealable elastomeric material.

(102) When disposable assembly **800** is filled with medicament but the pump is not yet dispensing, trocar seal assembly **840** acts as a medicament seal or a primary packaging seal. While trocar **847** is in place within drum **843** or the integrated drum-septum body in the first (or “seal”) position described above, trocar seals **845** engage trocar outer surface **866** to seal the interface between trocar **847** and drum **843** or integrated body to seal the fluid path through drum **843** (or combined drum/septum). As in earlier exemplary embodiments, trocar **847** may be made of stainless steel, or similar biocompatible material, and is used to penetrate the skin and a short distance into the flesh, to make a channel for cannula **841**, which may be made of TEFLON®/PTFE or other biocompatible material. In at least some instances, the associated pump will be pressed against the patient's skin, thereby driving the cannula **841** and trocar **847** below the skin surface. Once cannula **841** is in place, the trocar may be retracted to a second (or “unseal”) position to permit fluid flow from the reservoir outlet to the cannula **841**. For example, trocar **847** may be retracted into the pump **100/100a**, or removed from pump **100/100a** and the remainder of the trocar seal assembly **840** through septum **846** or (a combined drum/septum), to permit fluid flow. Septum **846** (or the septum portion of a combined drum/septum) will then seal end of drum outlet **864** opposite the cannula **841**.

(103) Another type of seal is a seal at a fill port where medicament is introduced into the reservoir. In some cases, it may be desirable to prevent medicament from passing from a reservoir **636/836/936** through a reservoir connection **639/739/839/939** until dispensing begins, and this may be achieved with fill plug seal assembly **904/1004** in the fill port **415/515/815/915**. As can be seen in the exemplary embodiment of FIGS. 15A-17E, a reservoir **936** may be filled with medicament through fill port **915**. To prevent medicament from traveling from reservoir **936** through reservoir connection **939** to an attached reservoir outlet and cannula (not shown) before dispensing begins, a fill plug **910/1010** of fill plug seal assembly **904/1004** may be inserted into fill port **915** to block passage of medicament from reservoir **936** to reservoir connection **939**. For instance, if it is preferable, during storage and/or before dispensing, to prevent medicament from being positioned in reservoir connection **639/739/839/939** and drum inlet **649/749/849**, as may be the case in the

embodiments above, and instead to maintain all medicament within reservoir **636/736/836/936** during storage and/or before dispensing, the following embodiments, or their equivalent, may be employed instead or in addition to the embodiments above.

(104) In one exemplary embodiment of a seal at the fill plug, a fill plug seal assembly **904** includes a thin seal **980**, such as a thin foil or other material that is impermeable and compatible with the medicament contained in the reservoir when in an intact state, is used to seal reservoir **936**, as shown in FIGS. **16A-16D**. A fill plug **910**, made of rubber or other relatively conformable sealing material, is positioned in fill port **915**. Fill plug **910** includes an inner surface that defines an inlet lumen **949**. A core pin **940** may be positioned through the inlet lumen **949** of fill plug **910**, as shown in FIGS. **16B-16D**. Core pin **940** may have a sharp tip **944** and may be attached to a push button **950** (FIGS. **16C** and **16D**), which may or may not be removable from core pin **940**. A fill plug outlet lumen **964** connects the inlet lumen **949** to the reservoir connection **939**.

(105) When device **100/100a** is ready for use, and the patient wants medicament to be dispensed from reservoir **936**, push button **950** should be pressed down until it is fully seated in fill plug **910**. As button **950** is pressed, core pin **940** advances from a first (or “seal”) position towards reservoir **936**, and sharp tip **944** penetrates seal **980** at a second (or “unseal”) position, allowing medicament to flow from reservoir **936**, through fill plug **910** of fill plug seal assembly **904** (by way of lumens **949** and **964**), and through reservoir connection **939**. Fill plug **910** includes an outer surface **962** with seals **912** that engage the inner surface **960** of fill port **915** to prevent medicament from leaking out around the outside of fill plug **910**. Core pin seals **942** prevent medicament from leaking through the fill plug and around the outside of core pin **940** after sharp tip **944** has penetrated seal **980**. Plug seals **912** and core pin seals **942** may be made of rubber or other relatively conformable sealing material, such as by injection molding the fill plug **910**, plug seals **912** and core pin seal **942** in one piece. Core pin **940** may be made of stainless steel, PEEK, ceramic or other relatively rigid biocompatible material, while push button **950** may be made of COP, Delrin, polycarbonate, or the like.

(106) Another exemplary embodiment of primary packaging at the fill plug is shown in FIGS. **17A-17E**. In this example, fill plug **1010** and core pin **1040** of fill plug seal assembly **1004** seal the reservoir while in storage and/or before dispensing. More specifically, fill plug **1010** includes an inner surface that defines an inlet lumen **1049**, and core pin **1040** is positioned within the inlet lumen. A fill plug outlet lumen **1064** connects inlet lumen **949** to reservoir connection **1039**. When core pin **1040** is in the storage or pre-dispensing position (i.e., the first (or “seal”) position where the button **1050** has not be pushed yet), the core end seal **1046** engages the inner surface of fill plug **1010** to seal the passage from reservoir **936** through fill plug **1010** to reservoir connection **939** (which is defined by lumens **1049** and **1064**), as best seen in FIG. **17C**. When device **100/100a** is ready for use, and the patient wants medicament to be dispensed from reservoir **936**, push button **1050** should be pressed down until it is fully seated in fill plug **1010**. As best seen in FIG. **17E**, as button **1050** is pressed, core pin **1040** advances towards reservoir **936** to a second (or “unseal”) position where core end seal **1046** disengages from the inner surface of fill plug **1010** that defines inlet lumen **1049**, leaving a gap **1055** through which medicament may flow from reservoir **936**, around core end seal **1046**, through lumens **1049** and **1064** of fill plug **1010**, and through reservoir connection **939**. Plug seals **1012** on the outer surface of fill plug **1010** prevent medicament from leaking out around the outside of fill plug **1010**, while core pin seals **1042** prevent medicament from leaking through the fill plug **1010** and around the outside of core pin **1040** after core end seal **1046** has been disengaged from fill plug **1010**. As above, fill plug **1010**, plug seals **1012**, and core pin seals **1042** may be made of rubber or other relatively conformable sealing material, such as by injection molding; core pin **1040** may be made of COP, Delrin, polycarbonate or other biocompatible polymeric material; and push button **1050** may be made of COP, Delrin, polycarbonate, or the like. While FIGS. **17A-17E** show core pin **1040** with a blunt end **1044**, this exemplary embodiment and the directly preceding exemplary embodiment may be combined by

adding a thin seal **980** to the end of reservoir **936**, and a sharp tip like sharp tip **944** to blunt end **1044**, so that pushing button **1050** punctures thin seal **940** as well as creating gap **1055** around the core end seal **1046**.

(107) To prevent accidental depression of button **1050** (or button **950**), a button guard may be inserted between button **1050/950** and fill plug **1010/910**. An exemplary button guard **1160** is shown in FIGS. **18A-18C**. When button guard **1160** is in place, it is not possible to press button **950/1050** to move core pin **940/1040** to break the seal **980/1046** to allow fluid to flow through fill plug **910/1010**. Button guard **1160** may be made of COP, Delrin, polycarbonate or other similar polymeric material.

(108) Various methodologies and systems are presented here in the context of the exemplary structures described in the preceding sections, and illustrated in the various figures, for the purpose of explanation only. Although the present methodologies and systems may employ the structures described above, they are not limited thereto. Additionally, embodiments of the present inventions may incorporate any one, combinations of less than all, or all of the methodologies or devices referenced above.

(109) Although the inventions disclosed herein have been described in terms of the preferred embodiments above, numerous modifications and/or additions to the above-described preferred embodiments would be readily apparent to one skilled in the art. By way of example, but not limitation, the present medicament sealing assemblies may be incorporated into fully disposable infusion pumps. It is intended that the scope of the present inventions extends to all such modifications and/or additions and that the scope of the present inventions is limited solely by the claims set forth below or later added.

(110) Finally, with respect to terminology that may be used herein, whether in the description or the claims, the following should be noted. The terms “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” and the like are open-ended and mean “including but not limited to.” Ordinal terms such as “first,” “second,” “third,” do not, in and of themselves, connote any priority, precedence, or order of one element over another or temporal order in which steps of a method are performed. Instead, such terms are merely labels to distinguish one element having a certain name from another element having a same name (but for the ordinal term) to distinguish the elements. “And/or” means that the listed items are alternatives, but the alternatives also include any combination of the listed items. The terms “approximately,” “about,” “substantially” and “generally” allow for a certain amount of variation from any exact dimensions, measurements, and arrangements, and should be understood within the context of the description and operation of the invention as disclosed herein. Terms such as “top,” “bottom,” “above,” and “below” are terms of convenience that denote the spatial relationships of parts relative to each other rather than to any specific spatial or gravitational orientation. Thus, the terms are intended to encompass an assembly of component parts regardless of whether the assembly is oriented in the particular orientation shown in the drawings and described in the specification, upside down from that orientation, or any other rotational variation therefrom.

Claims

1. An apparatus for a fluid delivery system, the apparatus comprising: a medicament reservoir; a reservoir outlet configured for fluid communication with the medicament reservoir; and a trocar seal assembly integrated into the apparatus and associated with the reservoir outlet, the trocar seal assembly comprising: a drum having a drum lumen extending therethrough; an inlet configured for fluid communication with the reservoir outlet; a cannula having a lumen configured for fluid communication with the inlet; and a trocar with a rod and a sharp end that is movable through the drum lumen and relative to the reservoir outlet between a first position in which the trocar is at least partially disposed within the cannula lumen and a second position in which the trocar is not

disposed within the cannula lumen.

2. The apparatus of claim 1, wherein, when the trocar is in the first position, the trocar limits medicament flow through the reservoir outlet to the cannula lumen, and when the trocar is in the second position, the trocar permits medicament flow from the reservoir outlet to the cannula lumen.
 3. The apparatus of claim 1, wherein the trocar seal assembly further comprises a cannula insert at least partially disposed within the cannula lumen, and wherein the trocar is moveable through the cannula insert.
 4. The apparatus of claim 3, wherein the cannula insert is configured to guide the trocar into place within the cannula lumen and prevent the sharp end of the trocar from piercing the cannula.
 5. The apparatus of claim 1, wherein the trocar seal assembly comprises a septum.
 6. The apparatus of claim 5, wherein the septum is adjacent to the drum or integral with the drum.
 7. The apparatus of claim 1, wherein the trocar is removable from the remainder of the trocar seal assembly.
 8. An infusion pump comprising: the apparatus of claim 1; and a fluid displacement device that drives medicament from the reservoir.
 9. The infusion pump of claim 8, wherein the fluid displacement device includes a plunger within the reservoir and a drive mechanism that drives the plunger.
 10. The infusion pump of claim 9, wherein the drive mechanism comprises a motor.
 11. An apparatus for a fluid delivery system, the apparatus comprising: a reservoir configured to retain fluid therein; a reservoir outlet in fluid communication with the reservoir; and a seal assembly comprising: a seal having a seal lumen extending therethrough; an inlet configured for fluid communication with the reservoir outlet; a cannula having a cannula lumen configured for fluid communication with the inlet; and an elongate rod that is movable through the seal lumen between a first position and a second position, wherein: in the first position, the elongate rod is at least partially disposed within the cannula lumen such that the elongate rod limits fluid flow from the inlet into the cannula lumen, and in the second position, the elongate rod is not disposed within the cannula lumen such that fluid flow is permitted from the inlet into the cannula lumen.
 12. The apparatus of claim 11, wherein the second position of the elongate rod is axially retracted relative to the first position of the elongate rod.
 13. The apparatus of claim 11, wherein the seal assembly further comprises a cannula insert at least partially disposed within the cannula lumen, and wherein the elongate rod is moveable through the cannula insert.
 14. The apparatus of claim 13, wherein the cannula insert is configured to guide the elongate rod into place within the cannula lumen and prevent a distal end of the elongate rod from piercing the cannula.
 15. The apparatus of claim 11, wherein the seal assembly comprises a septum.
 16. The apparatus of claim 15, wherein the septum is adjacent to the seal or integral with the seal.
 17. The apparatus of claim 11, wherein the elongate rod is removable from the remainder of the seal assembly.
 18. An infusion pump comprising: the apparatus of claim 11; and a fluid displacement device that drives fluid from the reservoir.
 19. The infusion pump of claim 18, wherein the fluid displacement device includes a plunger within the reservoir and a drive mechanism that drives the plunger.
 20. The infusion pump of claim 19, wherein the drive mechanism comprises a motor.
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